

What a Motor Prior Is Worth: A Frozen Body Model and a Teacher Compared to Convergence and Through Withdrawal

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Abstract—Sensorimotor theory offers two accounts of what a learner brings to a motor task: an internal forward model of the body, and guidance toward the correct response. We instantiate both as priors for one soft actor-critic learner on two MyoSuite musculoskeletal bodies, a frozen body model learned from reward-free babbling and a frozen teacher acting through a distillation term, with twelve matched seeds, an adversarial control for each route, and training run until the arms stop moving. Three results follow. The body model helps where the body is hard: its advantage over model-free learning is a transient on a one-joint elbow and stable on a four-joint finger. Guidance is only as good as its teacher: it keeps a small advantage on the elbow, whose teacher is competent, while on the finger, whose teacher barely moves, a student held near it ends worse than one given no aid. And withdrawing either prior costs no dependence: with the learner’s real-data signal held fixed, losing the body model costs 22 mrad that has dissipated by convergence, and swapping the teacher for a zero-information snapshot of the student shows that the immediate drop after withdrawal is the deletion of a loss term, that no component grows with attachment across seven durations and two coaching regimes, and that what persists is the loss of the teacher’s information. A fixed early endpoint reverses the first two verdicts; matching the two priors’ data budgets changes neither.

Index Terms—Developmental robotics, sensorimotor learning, internal models, guidance hypothesis, teacher–student distillation, reinforcement learning, musculoskeletal control.

I. INTRODUCTION

HUMANS and animals acquire motor skills in a handful of trials; deep reinforcement learning on muscle-actuated bodies can need hundreds of millions of environment steps. Computational sensorimotor theory offers two accounts of what a biological learner brings to a task that an agent trained from scratch does not. One is an internal forward model of the body, predicting the sensory consequences of motor commands [1], [2] and adapted in the intrinsic coordinates of joints and muscles [3]. The other is guidance: the guidance hypothesis [4], [5], [6] holds that augmented feedback steers the learner toward the correct response while breeding a dependence on it, so that the longer it is given the larger the fall when it is withdrawn. Both are claims about a *prior* in the developmental sense, something present

before task learning begins, and both can be instantiated for an artificial learner and compared on equal terms.

We do that here. Under one soft actor-critic backbone on two MyoSuite bodies, a one-joint elbow and a four-joint finger, we add either a frozen forward model of the body, learned once from reward-free motor babbling and used for short-horizon imagined rollouts, or a frozen teacher policy acting through a distillation term. Each route is given the adversarial control the other has: a random-initialised model in the same imagination machinery, a random-initialised teacher in the same distillation loss. Every arm has twelve matched seeds, every comparison is a paired contrast on the seed, and training runs until the arms have stopped moving. We propose no algorithm. The claim is about what a motor prior is worth, and it has three parts.

The body model helps where the body is hard. Its advantage over model-free learning is a transient of the first 12k steps on the elbow, not detected at 100k, and a stable 40 mm on the finger, detected at every checkpoint past 14k.

Guidance is only as good as its teacher. On the elbow, whose teacher is competent, the coached student keeps a small advantage at convergence. On the finger, whose teacher is within 3.2 mm of doing nothing, a student held near it ends 28 mm worse than one given no aid, and worse than one held near a random teacher.

Withdrawing either prior costs no dependence. Withdrawal as the literature performs it changes two things at once. Purging the body model’s imagined buffer also doubles the learner’s real-data gradient signal; with that signal held fixed the loss costs +22 mrad on the elbow at 12k, where the standard protocol detects nothing, and the cost has dissipated by 100k. Zeroing a teacher’s distillation weight deletes a loss term as well as the teacher’s information; replacing the teacher by a frozen snapshot of the student itself, so that the term survives but carries nothing, shows that the immediate drop is the loss term, that across seven attachment durations and two coaching regimes no component grows with attachment, and that what persists at convergence is about 8 mrad of genuinely lost teacher information.

Two checks close the case. Scored at 12k, as an earlier version of this study scored it, the first two verdicts read the other way on both bodies: that endpoint sat inside the only run of consecutive checkpoints at which the elbow benefit was visible, and inside the only dip in which the finger benefit was

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not. And the body model saw $3.3\times$ to $6.7\times$ more transitions than the teacher; cut to the teacher’s budget, its endpoint changes by $+1.6$ mrad and $+2.8$ mm, neither detected. The runs, analysis scripts, and figures are released with the paper.

II. BACKGROUND AND RELATED WORK

The guidance hypothesis. Salmoni, Schmidt and Walter [4] proposed that frequent or immediate augmented feedback guides the learner to the correct response during acquisition while preventing the processing that supports retention; when the feedback is removed, performance falls. Schmidt and Bjork [5] generalised the account to practice conditions at large, and Winstein, Pohl and Lewthwaite [6] showed the same pattern for physical guidance. Its best-known test [7] supports the retention limb more clearly than the acquisition limb, and the reversal is weak across studies [8]. Two features matter here. It is a claim about *dependence*: the learner is worse after withdrawal than one who never had the guidance, or than one who had less of it. And its mechanism is that guidance *substitutes* for the learner’s own error detection and correction, so that the learner does not practise them.

Teacher–student learning in robotics. Privileged learning [9], [10] trains a teacher with access to state the deployed policy will not have, then distils it into a student through a behaviour cloning loss on the student’s own observations. In current legged and humanoid pipelines [11] the distillation stage may be the entire training of the student, with no reward term; where a reward term is present the distillation loss is auxiliary and is annealed or cut. In both cases the post-withdrawal drop is used as a diagnostic, and the interpretation offered is the guidance hypothesis’ one, usually without naming it. We found no work in reinforcement learning or robotics that runs a control for the loss term. A policy initialised by behaviour cloning and then fine-tuned by off-policy reinforcement often collapses at the transition, because the critic’s errors on out-of-distribution actions drive the actor away from the cloned behaviour [12], [13], [14]; Sec. IV-D shows that this transition, not the teacher, is what grows with attachment when guidance replaces the learner’s own signal.

Model-based RL and body models. World Models [15] and the Dreamer series [16], [17], [18] optimise a policy on imagined rollouts while updating the model online; MBPO [19] bounds compounding error with short rollouts branched from real data, the basis for our Dyna step; PETS [20] and TD-MPC2 [21] keep the model learning throughout. Robot self-modelling [22] infers a body model from exploratory actions and goal babbling [23] learns an inverse model by directed exploration, where we use undirected babbling to learn a forward model and freeze it. Plan2Explore [24] and APV [25] pretrain task-agnostic models in a reward-free phase; MOREL, MOPO and COMBO [26], [27], [28] freeze a pretrained model and optimise under it with pessimism, where we use a babble-learned model optimistically and hand the arm the environment’s analytic reward at imagined states. Mainstream muscle RL is model-free [29], [30], [31], [32]; MuscleVAE [33] and FreeMusco [34] learn world models for muscle control but co-learn them during skill training. MyoSuite [35] provides the bodies.

III. METHOD

A. Bodies, task and backbone

`myoElbowPose1D6M` is a one-degree-of-freedom elbow driven by six muscles; error is a joint-angle error in milliradians (mrad). `myoFingerReachRandom-v0` is a four-joint index finger driven by five muscles reaching a three-dimensional fingertip target; error is a Euclidean fingertip error in millimetres (mm). The two are never compared in magnitude. The task is reach; episodes are 100 steps from a fixed start pose; training draws from eight fixed targets; evaluation is greedy on twenty disjoint held-out targets, so every number here is a generalisation measurement. One soft actor-critic [36] backbone is used in every condition: two hidden layers of 128 units, update-to-data ratio 2, batch 128, 1,000 warm-up steps, learning rate 3×10^{-4} , automatic entropy temperature. Conditions differ only in what is added.

B. The two priors

The *forward-model* route adds a body model $f_\theta(s, a) \mapsto s'$ trained once on task-agnostic, reward-free Ornstein–Uhlenbeck muscle activations (10^5 transitions on the elbow, 2×10^5 on the finger), then frozen, and used for short-horizon Dyna [37] rollouts branched from sampled real states; the environment’s analytic reward is evaluated at the imagined states, so every reading of this arm is scoped to a forward model *plus* a known reward function. Each update draws 64 real and 64 imagined transitions where model-free arms draw 128 real.

The *guidance* route adds a frozen teacher actor trained for 30k rewarded steps on the same body. The student’s actor loss is

$$\mathcal{L}_\pi = q \mathcal{L}_{\text{SAC}} + c \|\tanh \mu_{\text{student}}(o) - \tanh \mu_{\text{teacher}}(o)\|^2,$$

on replayed observations o , with $c = 1$ while the teacher is attached; the critic trains on reward in every arm. In the *additive* regime ($q = 1$ always) the term supplements the student’s own policy gradient, as concurrent augmented feedback does; this is the coach used everywhere unless stated. In the *replacing* regime ($q = 0$ while attached, $q = 1$ from withdrawal) the actor is moved only by the teacher until withdrawal, as physical guidance moves the limb and as reward-free distillation [11] does.

C. Arms

Adversarial controls. `randprior` replaces the babble-trained model with a random-initialised frozen model in identical Dyna machinery; `randcoach` replaces the trained teacher with a random-initialised one distilled through the same loss at the same weight. `priorcoach` carries both aids.

Withdrawing the body model at step 8000: `keep` never withdraws; `soft` stops generating imagined transitions but leaves the imagined buffer in place, so updates keep drawing 64 real and 64 stale imagined; `purge` also empties the buffer, after which every batch is 128 real, the strict withdrawal of the original protocol, which removes imagination and doubles distinct real transitions per update at once; `matched`

withdraws as *purge* does but draws 64 real transitions and repeats them, holding distinct real transitions per update at 64. Real-data content is therefore 64 in *keep*, *soft* and *matched* and 128 in *purge*, so only contrasts among the first three are single-factor.

Withdrawing the teacher at step t_w : *none* (no teacher); *constant* ($c = 1$ throughout); *abrupt* ($c \rightarrow 0$ at t_w , the literature’s withdrawal); *selfanchor* (at t_w the teacher is replaced by a frozen copy of the student’s own actor, c unchanged: the term survives with the same weight and form but carries no external information); *randanchor* (replaced by a freshly initialised actor); and, in the replacing regime only, *gateon* ($q \rightarrow 1$ at t_w with the real teacher kept). Three paired contrasts decompose the measured cost: *abrupt* – *constant* is losing the teacher *and* the loss term, *selfanchor* – *constant* is losing the teacher only, and *abrupt* – *selfanchor* is losing the loss term only. The second is where dependence would have to show up; the third is the confound. Snapshots are taken of the student’s actor on the student’s observations, with any observation normaliser frozen with them.

D. Metrics and statistics

Endpoint: held-out error at the end of training. *Dip*: error at $t_w + 1000$ minus error at t_w . *Post-4k*: error 4,000 steps after t_w , comparable across t_w . *Attachment trend*: per-seed ordinary-least-squares slope of a component against t_w , then a paired- t interval across seeds, per 1,000 attached steps. The seed is the unit of analysis and every arm has $n = 12$ matched seeds; every comparison is a direct paired contrast on the seed, never an inference from two separate verdicts; intervals are paired- t 95% intervals with $df = 11$; an interval excluding zero is called significant and marked *; a non-significant cell is a failure to detect at this power, never evidence of equivalence.

E. Reproduction

Every arm of the converged grid reproduces the original study’s 12k arm means seed for seed at its 12k checkpoint, the withdrawal arms reproduce its kept, soft-withdrawn and strict-withdrawn arms seed for seed, and the teacher-side arms reproduce three published cells to the digit: *myoElbow abrupt* – *constant* endpoint $+16.00 [+6.49, +25.52]$, *myoElbow body-model-versus-coach* $+10.89 [+2.52, +19.26]$, *myoFinger abrupt* – *constant* $+46.37 [+22.88, +69.87]$. New arms are measured on the same footing as the old ones.

IV. RESULTS

A. The body decides between the routes

Table I and Fig. 1 score the six core arms at 100,000 steps. On *myoElbow* every arm but the random-model control has stopped moving: over the last 20,000 steps *blank*, *prior* and *coach* change by $+2.7$, -2.9 and -1.7 mrad against across-seed SDs of 11.3, 9.4 and 3.5; *randprior* is still falling (-25.0 , SD 29.6). The forward-model route’s advantage over model-free learning, -20.72 mrad $[-35.8, -5.6]$ at 12k, is $-4.09 [-11.11, +2.94]$ at 100k and is not detected; across fifty checkpoints it is significant at seven:

8k, 10k and 12k, and four isolated later ones with no two adjacent. The guidance route’s advantage shrinks from $-31.61 [-48.6, -14.6]$ to $-8.06 [-15.83, -0.28]$ but is still detected, at 43 of 50 checkpoints. The coach’s lead over the body model, $-10.89 [-19.3, -2.5]$ at 12k, decays to $-3.97 [-8.47, +0.53]$ and is not detected at the endpoint, though it is at 38 of 50 checkpoints including 96k; it does not reverse on this body within the budget. The content of each aid matters: the babble-trained model beats the random one by $-20.29 [-32.00, -8.59]$ and the trained teacher beats the random one by $-17.69 [-33.70, -1.68]$; the random model is itself worse than model-free learning ($+16.21 [+6.82, +25.59]$), the random teacher not distinguishable from it ($+9.63 [-8.07, +27.34]$). Combining the aids adds nothing detectable: *priorcoach* is $-5.24 [-12.93, +2.45]$ against model-free learning and $+2.81 [-1.02, +6.65]$ against the coach alone.

On *myoFinger* the same budget moves the verdicts the other way. The body model’s advantage over model-free learning, not detected at 12k ($-27.10 [-66.3, +12.1]$), is $-40.29 [-57.95, -22.64]$ at 100k and is detected at every checkpoint from 14k onward, 45 of 50 in all; the 12k endpoint sat in the one dip in which it was not. The guidance route does not improve past 12k on this body (122.77 mm at 12k, 123.00 at 100k), so the ordering of the two routes, already reversed at 12k ($+32.27 [+16.5, +48.0]$ for coach – prior), widens to $+68.63 [+59.19, +78.07]$ and holds at every checkpoint from 12k on. From 40k on the constant-weight coach is significantly worse than model-free learning at 27 of 31 checkpoints, $+28.33 [+16.11, +40.56]$ at 100k. The finger teacher as captured sits within 3.2 mm of the no-op referent (Sec. VI); a student held near it by a constant pull is held near a policy that barely moves. The random-model control holds ($-48.71 [-62.32, -35.09]$), the random model not distinguishable from model-free learning at $+8.41 [-14.17, +30.99]$; the random-teacher control reverses: at 12k the trained teacher beats the random one by -19.25 , at 100k the random one is better by $+13.86 [+7.62, +20.09]$, and is itself worse than model-free learning ($+14.48 [+0.74, +28.21]$). Combining the aids inherits the coach’s ceiling: *priorcoach* finishes $-8.38 [-15.57, -1.19]$ better than the coach alone, $+60.25 [+51.14, +69.36]$ worse than the body model alone and $+19.96 [+5.74, +34.17]$ worse than no aid.

The 12k endpoint was therefore not a neutral place to score. It overstated the body model on the elbow, where the only run of checkpoints at which its benefit was detectable was 8k to 12k, and understated it on the finger, where its benefit was detectable everywhere except a dip that contained 12k. The coach’s ordering relative to the body model is body-dependent, and on both bodies the direction it takes by 12k is the direction it keeps.

Budget matching. The body model saw $3.3\times$ (elbow) to $6.7\times$ (finger) more transitions than the teacher. We rebuilt the finger prior from exactly 30k babble transitions with the same recipe and reused a 30k elbow prior from the original budget sweep. Table II shows that cutting the budget changes the endpoint by $+1.55 [-0.87, +3.97]$ mrad and $+2.75 [-12.77, +18.27]$ mm, although the finger model’s

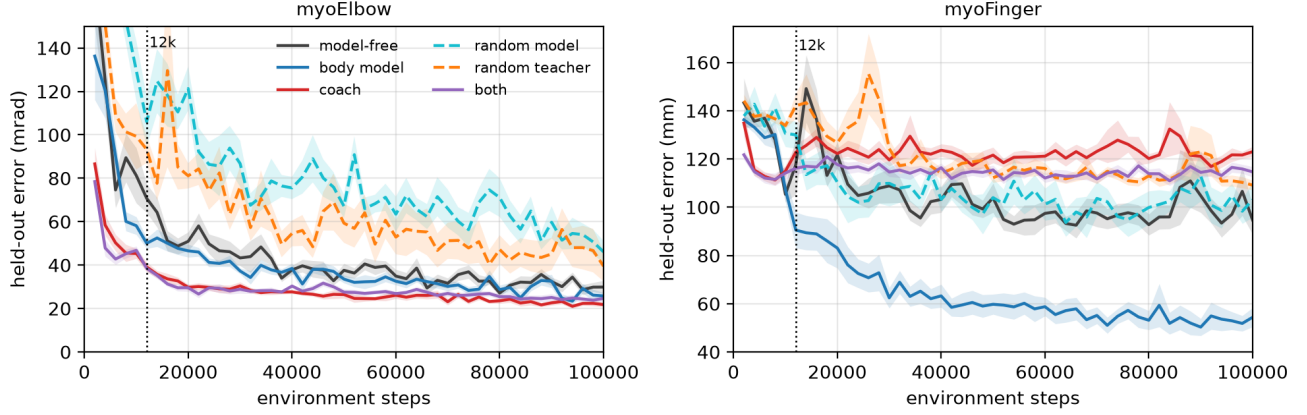


Fig. 1. Held-out error of the six core arms over 100,000 steps on myoElbow (left, mrad; axis clipped at 150) and myoFinger (right, mm): mean across 12 seeds, band ± 1 SE, dashed lines the random controls. The dotted line marks the 12k endpoint at which the original study scored every arm.

TABLE I

THE SIX CORE ARMS AT A CONVERGED BUDGET OF 100,000 STEPS, HELD-OUT, $n = 12$. ARM MEANS AT 12K AND 100K, THEN PAIRED CONTRASTS AT 100K WITH 95% INTERVALS; NEGATIVE FAVOURS THE FIRST-NAMED ARM. THE 12K CHECKPOINTS REPRODUCE THE ORIGINAL STUDY'S ARM MEANS SEED FOR SEED (RANDPRIOR WAS SCORED ON TRAINING TARGETS THERE).

	myoElbow (mrad)		myoFinger (mm)	
	12k	100k	12k	100k
blank	70.73	29.87	117.60	94.66
prior	50.02	25.78	90.50	54.37
randprior	106.03	46.08	129.98	103.08
coach	39.13	21.81	122.77	123.00
randcoach	93.26	39.50	142.02	109.14
priorcoach	38.49	24.63	116.13	114.62
coach-prior	-10.89	-3.97 [-8.47, +0.53]	+32.27	+68.63 [+59.19, +78.07]
prior-blank	-20.72	-4.09 [-11.11, +2.94]	-27.10	-40.29 [-57.95, -22.64]
coach-blank	-31.61	-8.06 [-15.83, -0.28]	+5.17	+28.33 [+16.11, +40.56]
priorcoach-blank	-32.25	-5.24 [-12.93, +2.45]	-1.47	+19.96 [+5.74, +34.17]
priorcoach-coach	-0.64	+2.81 [-1.02, +6.65]	-6.64	-8.38 [-15.57, -1.19]
prior-randprior	-56.01	-20.29 [-32.00, -8.59]	-39.48	-48.71 [-62.32, -35.09]
coach-randcoach	-54.14	-17.69 [-33.70, -1.68]	-19.25	+13.86 [+7.62, +20.09]

TABLE II

BUDGET MATCHING AT 12K. ENDPOINT MEAN (SEED S.D.) AND PAIRED CONTRASTS.

	myoElbow (mrad)	myoFinger (mm)
none	70.73 (22.77)	117.60 (43.03)
coach, 30k rewarded steps	39.13 (7.77)	122.77 (12.18)
prior, 30k babble	51.57 (8.73)	93.25 (22.02)
prior, full budget	50.02 (8.18)	90.50 (25.57)
prior _{30k} - prior _{full}	+1.55 [-0.87, +3.97]	+2.75 [-12.77, +18.27]
prior _{full} - coach	+10.89 [+2.52, +19.26]*	-32.27 [-48.05, -16.49]*
prior _{30k} - coach	+12.44 [+3.27, +21.61]*	-29.52 [-43.17, -15.87]*

five-step prediction error rises from 14.3 to 25.1 mm; the finger reversal survives matching. Matching raw transition counts is not matching information: the teacher's 30k steps are rewarded and on-task, the prior's are not, and the prior arm holds the analytic reward at imagined states.

B. Withdrawing the body model costs a transient

Table III gives the four arms at 12k. Measured as the original protocol measured it, withdrawing the body

model has no detectable cost on myoElbow: purge - keep is $+7.17 [-10.95, +25.29]$. With distinct real transitions per update held at 64, the same withdrawal is significantly costly: matched - keep is $+22.11 [+4.94, +39.28]$, and the change across the cut itself (8000 to 10000) is $+21.47 [+4.54, +38.40]$. The two protocols reach different verdicts because the strict withdrawal doubles the agent's real-data gradient signal at the moment it loses the model, and that gain offsets part of the loss; the direct contrast purge - matched is $-14.94 [-44.62, +14.73]$ and not significant, so we do not put a number on the offset. The soft arm locates the cost: with real-data content equal, keeping the stale imagined transitions rather than deleting them is worth $-20.02 [-37.34, -2.70]$ (soft - matched), while keeping them stale rather than fresh costs $+2.09 [-2.85, +7.04]$, not detected. By 8000 steps what the frozen body model is worth on the elbow is carried by the imagined transitions already in the buffer. On myoFinger neither protocol detects a cost (purge - keep $+14.19 [-0.95, +29.33]$; matched - keep $+9.97 [-1.50, +21.45]$), so this is an elbow result.

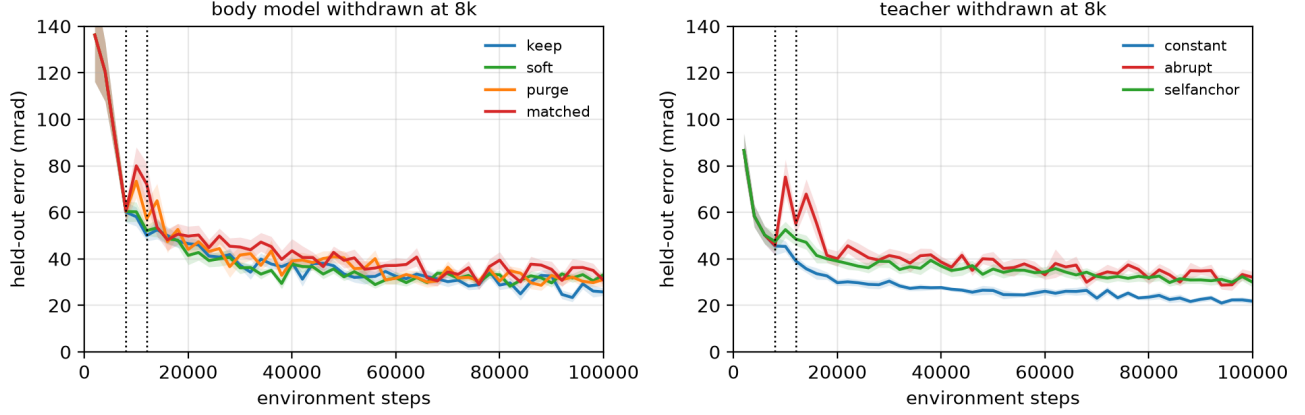


Fig. 2. Both withdrawals on myoElbow run to 100,000 steps (mean across 12 seeds, band ± 1 SE; dotted lines at the withdrawal step 8000 and at 12k). Left: the four body-model arms. Right: the three teacher arms. The body-model arms converge on one another; the two withdrawn-teacher arms stay above constant.

TABLE III

WITHDRAWAL OF THE BODY MODEL AT STEP 8000, SCORED AT 12000, HELD-OUT, $n = 12$. MEANS, THEN PAIRED CONTRASTS WITH 95% INTERVALS; POSITIVE MEANS WITHDRAWAL MADE THE ARM WORSE. †: SINGLE-FACTOR (REAL-DATA CONTENT EQUAL ON BOTH SIDES).

	myoElbow (mrad)	myoFinger (mm)
keep	50.02	90.50
soft	52.11	88.11
purge	57.19	104.69
matched	72.13	100.48
purge-keep	+7.17 [-10.95, +25.29]	+14.19 [-0.95, +29.33]
soft-keep†	+2.09 [-2.85, +7.04]	-2.39 [-13.74, +8.97]
matched-keep†	+22.11 [+4.94, +39.28]	+9.97 [-1.50, +21.45]
soft-matched†	-20.02 [-37.34, -2.70]	-12.36 [-24.84, +0.12]
purge-matched	-14.94 [-44.62, +14.73]	+4.22 [-12.76, +21.19]

It is also a transient. Table VI and Fig. 2 run the same arms and seeds to 100,000 steps; their 12k checkpoints reproduce the 12k runs seed for seed. By 100k the three withdrawn arms sit within 5–7 mrad of keep (25.78) and within 2.4 of one another: matched – keep is +5.05 [-0.98, +11.08], no longer detected, and soft – matched is +2.35 [-4.42, +9.11]. The one contrast detected at 100k runs the other way from 12k: keeping the stale buffer is now the worst of the four arms, soft – keep +7.40 [+2.72, +12.08]. What the matched control showed at 12k is a statement about what the standard protocol misses in the window it scores, not a persistent cost; on this body, after enough training, a learner that lost its body model is close to one that kept it.

C. Withdrawing the teacher costs no dependence

At one withdrawal point. Table IV decomposes the withdrawal cost at $t_w = 8000$. The immediate dip is the loss term: 89% of it on the elbow and all of it on the finger. On the elbow a genuine cost of losing the teacher appears only at the endpoint (+9.42 [+3.13, +15.70]), so short-term and long-term attributions are opposite within one experiment. On the finger there is none: swapping in the zero-information anchor beats keeping the teacher (-11.29 [-20.65, -1.93]), selfanchor is the only arm in which all twelve seeds

TABLE IV

TEACHER WITHDRAWAL DECOMPOSED AT $t_w = 8000$, $n = 12$; * MARKS AN INTERVAL EXCLUDING ZERO. ELBOW IN MRAD, FINGER IN MM.

Contrast	myoElbow		myoFinger	
	dip	endpoint	dip	endpoint
abrupt – constant (total)	+45.17*	+16.00*	+13.86*	+46.37*
selfanchor – constant (teacher)	+4.95	+9.42*	-0.27	-11.29*
abrupt – selfanchor (loss term)	+40.22*	+6.59	+14.13*	+57.66*
randanchor – selfanchor	+47.44*	+20.66*	+20.13*	+17.20*
constant – none	–	-31.61*	–	+5.17

Seeds recovering pre-withdrawal error (of 12): elbow none 11, constant 7, abrupt 6, selfanchor 8, randanchor 5; finger 12, 6, 6, 12, 3.

recover their pre-withdrawal error, and the teacher is not helping at 12k (constant – none +5.17 [-26.49, +36.82]). The randanchor arm is worse than selfanchor on both bodies and both metrics, so the anchor’s content matters: selfanchor is not “any loss term” but a zero-information one pointing in the right direction, which is what a control for the loss term should be. The finger abrupt arm does not dip and recover: it is at 131 mm one thousand steps after withdrawal and 169 at 12k, with the entropy temperature unchanged across arms, a drift of the policy under the reward gradient once the regulariser is removed.

TABLE V

TEACHER-WITHDRAWAL COMPONENTS AGAINST ATTACHMENT DURATION t_w . SLOPE: PER-SEED OLS OVER t_w , PAIRED- t INTERVAL, PER 1,000 ATTACHED STEPS. ENDPOINT ROWS ARE NOT ON EQUAL FOOTING ACROSS t_w .

		2k	3k	4k	5k	6k	7k	8k	slope
<i>myoElbow (mrad)</i>									
dip	teacher	46.0*	22.0*	25.6*	1.5	2.7	7.8	5.0	-6.23 [-7.79, -4.68]*
	loss term	13.6	17.7*	37.9*	42.6	77.9*	24.3*	40.2*	+4.76 [-0.13, +9.65]
	total	59.5*	39.7*	63.6*	44.1*	80.6*	32.1*	45.2*	-1.47 [-5.79, +2.84]
post-4k	teacher	25.1*	22.1*	8.2	3.4	4.2	7.2	9.4*	-2.88 [-5.86, +0.10]
	loss term	9.0	61.9*	34.7*	26.5*	36.2	21.2*	6.6	-3.11 [-8.69, +2.46]
	total	34.1*	84.0*	42.9*	29.9*	40.3*	28.4*	16.0*	-6.00 [-10.60, -1.39]*
endpoint	teacher	27.9*	11.4*	5.6	2.4	6.4	7.0	9.4*	-2.26 [-4.84, +0.31]
	loss term	-2.8	17.9	34.7*	23.8*	19.1	22.4*	6.6	+0.77 [-2.82, +4.36]
<i>myoFinger (mm)</i>									
dip	teacher	-0.6	4.4	7.0	4.8	0.4	2.2	-0.3	-0.35 [-1.46, +0.75]
	loss term	7.3	4.6	8.0*	10.0	13.7*	20.7*	14.1*	+2.09 [+0.46, +3.72]*
	total	6.6	8.9*	15.0*	14.8*	14.1*	22.8*	13.9*	+1.74 [+0.25, +3.22]*
post-4k	teacher	7.5*	3.9	7.0*	4.0	3.5	-6.4*	-11.3*	-2.87 [-4.27, -1.47]*
	loss term	14.7*	7.1	14.3*	3.7	-0.8	30.4	57.7*	+5.74 [+1.49, +9.98]*
	total	22.2*	10.9*	21.3*	7.7	2.7	24.1	46.4*	+2.87 [-1.74, +7.47]
endpoint	teacher	-3.3	-4.8	-10.0*	-10.5*	-9.9*	-9.9*	-11.3*	-1.21 [-2.52, +0.10]
	loss term	2.6	9.6	-1.0	21.5	13.4	55.1*	57.7*	+9.67 [+5.54, +13.80]*

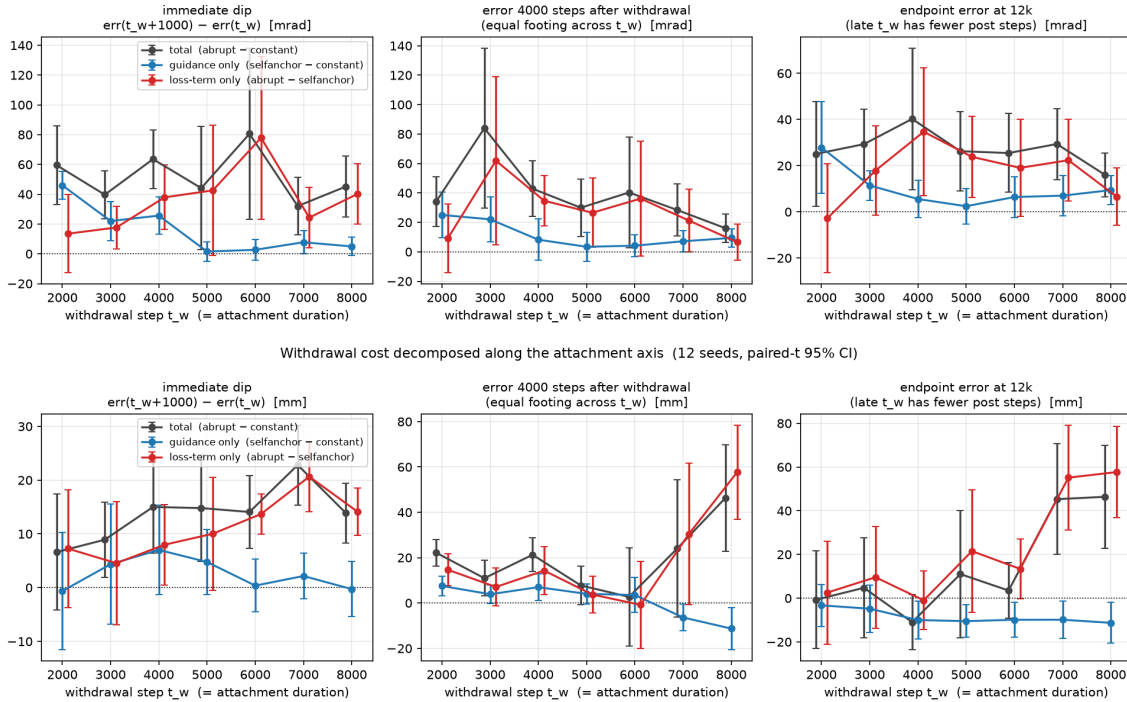
Withdrawal cost decomposed along the attachment axis (12 seeds, paired- t 95% CI)

Fig. 3. Teacher-withdrawal components against attachment duration on myoElbow (top, mrad) and myoFinger (bottom, mm): total (black), teacher only (blue), loss term only (red); $n = 12$, paired- t 95% intervals. Panels: immediate dip, error 4,000 steps after withdrawal, endpoint.

Along the attachment axis. The hypothesis predicts that the cost of losing the teacher grows with how long it was attached. We withdrew at every 1,000 steps from 2,000 to 8,000, running abrupt and selfanchor at each point with constant and none shared (Table V, Fig. 3). On myoElbow the teacher component of the dip is significant at $t_w \leq 4000$ and null from 5,000 on; the per-seed slope is $-6.23 [-7.79, -4.68]$ mrad per 1,000 attached steps and no seed deviates from the downward

direction. The component tracks how much the teacher is contributing at that moment (constant - none at each t_w : $-86, -90, -66, -55, -24, -25, -44$), not an accumulated reliance. The loss-term component does not decay (slope $+4.76 [-0.13, +9.65]$) and is large and noisy at $t_w = 6000$, where several abrupt seeds collapse; we claim that it does not decay, not that it grows. On myoFinger the total dip *does* grow with attachment (slope $+1.74 [+0.25, +3.22]$ mm

TABLE VI

BOTH WITHDRAWAL DECOMPOSITIONS ON MYOELBOW, WITHDRAWN AT 8000, SCORED AT 100,000, $n = 12$; THE 12K COLUMN REPEATS THE VALUES THESE SAME RUNS GIVE AT THEIR 12K CHECKPOINT. POSITIVE MEANS WITHDRAWAL MADE THE ARM WORSE. BODY-MODEL ARM MEANS AT 100K: KEEP 25.78, SOFT 33.18, PURGE 31.12, MATCHED 30.84.

	12k		100k
<i>body model</i>			
purge-keep	+7.17	+5.34	[-3.14, +13.81]
soft-keep	+2.09	+7.40	[+2.72, +12.08]
matched-keep	+22.11	+5.05	[-0.98, +11.08]
soft-matched	-20.02	+2.35	[-4.42, +9.11]
purge-matched	-14.94	+0.28	[-8.15, +8.72]
<i>teacher</i>			
abrupt-constant	+16.00	+10.29	[+3.96, +16.62]
selfanchor-constant	+9.42	+8.16	[+2.17, +14.14]
abrupt-selfanchor	+6.59	+2.13	[-3.64, +7.90]

per 1,000 steps), the textbook signature of the hypothesis; but the teacher component is null at every t_w (slope $-0.35 [-1.46, +0.75]$) and the growth is entirely the loss-term component (slope $+2.09 [+0.46, +3.72]$). At post-4k the teacher component turns negative late ($t_w = 7000$: -6.39^* ; 8000 : -11.29^* ; slope $-2.87 [-4.27, -1.47]$) and the loss-term slope is $+5.74 [+1.49, +9.98]$: the longer the teacher was attached, the better off the student is for having swapped it out, and the worse for having had the term deleted. A study measuring the total drop alone would report the hypothesis confirmed on this body.

At a converged budget. The lower half of Table VI and the right panel of Fig. 2 run the elbow decomposition at $t_w = 8000$ to 100,000 steps. Unlike the body model's, the teacher's cost is not a transient, and by 100k it is almost entirely the information component: abrupt – constant is $+10.29 [+3.96, +16.62]$, down from $+16.00$ at 12k but still detected; of it, losing the teacher's information accounts for $+8.16 [+2.17, +14.14]$ (selfanchor – constant), while the loss-term component, $+2.13 [-3.64, +7.90]$ (abrupt – selfanchor), is no longer detected. What survives training is the loss of a competent teacher, not the optimisation event of deleting the term, and it is still not a cost that grew with attachment.

D. When guidance replaces the learner's own signal

The additive coach is the wrong model of physical guidance, which substitutes for the learner's own error correction rather than supplementing it. The replacing regime is the right one, and it is also what reward-free distillation does in current humanoid pipelines [11]. We ran it on myoElbow at $t_w \in \{2000, 4000, 6000, 8000\}$ with constant, abrupt, selfanchor and gateon (Table VII). Without the gateon arm the first row confirms the hypothesis: the cost of losing the teacher is large at every t_w and grows with attachment (post-4k slope $+5.77 [+1.30, +10.25]$; endpoint slope $+8.66 [+2.99, +14.33]$), an order of magnitude above the additive regime at late t_w . With it the confirmation dissolves. Keeping the real teacher while switching on the student's own gradient produces the same collapse, seed by seed: selfanchor – gateon is within 3.1 mrad at every t_w , with

intervals as narrow as $[-2.5, +1.7]$, and no significant slope on any metric. What grows with attachment is the cost of first activating the learner's own policy gradient after a cloning-only phase, the known collapse when a behaviour-cloned policy is fine-tuned by off-policy reinforcement [12], [13], [14], and it is unrelated to the teacher. After a cloning-only phase the student's snapshot is the teacher, so selfanchor cannot price the teacher's information in this regime; gateon can, and finds none. The pure-distillation constant arm ends at 27.86 mrad, better than the additive constant (39.13), and abrupt in this regime, plain SAC after t_w , recovers in 0 of 12 seeds at $t_w = 4000$ and 6,000, ending at 133 and 168 mrad, worse than a student that never had a teacher (70.73).

V. DISCUSSION

What dependence would have to look like. The self-anchor control makes the notion measurable: dependence is the part of the withdrawal cost that survives replacing the teacher by a zero-information stand-in, and the guidance hypothesis predicts that this part grows with attachment. On two bodies, seven durations, and two regimes, it never does. In the additive regime it tracks the teacher's current contribution and fades as the learner becomes competent; on the finger, where the teacher contributes little at any time, it is null throughout and swapping the teacher out is better than keeping it. In the replacing regime the growing cost is real but belongs to the learner's own machinery, an actor that has never been moved by its critic, not to the teacher. What does persist on the elbow to a converged budget is about 8 mrad of genuinely lost teacher information, with the loss-term shock gone; that is the cost of no longer having a competent teacher, and it did not grow with how long the teacher was attached. The hypothesis' mechanism, that guidance prevents the learner from practising its own error correction [5], has a visible shadow here: a cloning-only actor pays for not having practised when its own gradient is switched on, and pays more the longer it went without. But the human prediction is about the guidance being missed, and that is what we do not find. The zero-information anchor has a natural human analogue, guidance that is present but carries no information about the target, and the question is now a precise one.

Guidance is only as good as its teacher. On the acquisition limb the coach does what the hypothesis says on both bodies through 8k. What happens after that is body-dependent in a way the hypothesis does not predict but does explain. On the elbow the teacher is competent and the student, held near it, plateaus near it, keeping a small advantage over model-free learning at convergence. On the finger the teacher is within 3.2 mm of doing nothing; a student held near it is held near a policy that does not move, by 40k it is worse than no aid, and at convergence it is worse than a student held near a random teacher. A constant pull toward a poor target is a ceiling, not a prior, and a body model that on its own beats model-free learning by 40 mm cannot pull a student out of it.

The body model helps where the body is hard. Internal-model theory would have a learned body model help most where the body is hardest to control without one, and that is the

TABLE VII

REPLACING REGIME, MYOELBOW. DIP IN MRAD AT EACH t_w , AND PER-SEED SLOPES OF THE POST-4K AND ENDPOINT METRICS. ENDPOINTS: CONSTANT 27.86; ABRUPT 107.7, 133.0, 168.2, 180.8; SELFANCHOR 45.9, 59.8, 110.8, 86.7.

	2k	4k	6k	8k	slope post-4k	slope endpoint
selfanchor – constant	58.7*	104.7*	109.4*	67.2*	+5.77 [+1.30, +10.25]*	+8.66 [+2.99, +14.33]*
gation – constant	59.1*	105.4*	110.2*	64.1*	+5.90 [+0.26, +11.54]*	+8.82 [+2.58, +15.06]*
selfanchor – gation	-0.4	-0.8	-0.8	+3.0	-0.13 [-3.44, +3.19]	-0.15 [-5.25, +4.95]
abrupt – selfanchor	54.4*	86.5*	58.6*	76.4*	+4.32 [-4.60, +13.24]	+4.06 [-1.82, +9.95]
abrupt – constant	113.1*	191.2*	168.0*	143.6*	+10.10 [+0.52, +19.68]*	+12.72 [+6.47, +18.98]*

direction the two bodies take: on the one-joint elbow a model-free learner closes the gap by 100k and the model’s benefit is a transient of the first 12k; on the four-joint finger it is present at convergence and at nearly every checkpoint before it. On the elbow what the model is worth at 8k is carried by the imagined transitions already stored, and deleting them costs 20 mrad in the window after the cut, which the original protocol could not see because it deleted them and doubled the real-data signal in the same step; by 100k that cost is gone and the arm that kept a stale buffer is the worst of the four. The body model’s value on this body is front-loaded and temporary, consistent with its endpoint advantage being a transient of the same window. The caveats are that the elbow’s generalisation test is weak, that two bodies are an ordering and not a law, and that the prior arm is handed the analytic reward at imagined states, which no biological reading can take for granted.

Measurement. Each of the three results needed a measurement the literature does not usually make. An endpoint fixed in advance and not verified as an asymptote landed inside a transient on both bodies; budgets of the order of 12k are common in muscle control, and the remedy is not a larger number but a check that the arms have stopped moving. A withdrawal that changes two things at once cannot be read as the cost of the thing withdrawn; the body-model side needed a matched-replay arm, the teacher side a self-anchor, and a cloning-only pipeline a *gation* arm as well, since a distillation learner with no reward term [11] freezes when its coefficient is zeroed and the swap is the only withdrawal that means anything. Fading schedules deserve the same decomposition: a taper spreads the loss-term deletion over time and may be reducing the shock rather than the dependence.

VI. LIMITATIONS

The study is single-task reach, simulation only, on deterministic and fully observed bodies, conditions that favour a frozen forward model. Two bodies are analysed; myoElbow has one degree of freedom, so its twenty held-out targets interpolate between eight training targets and its generalisation test is weak. One teacher per body: the finger teacher as captured scores 169.1 mm against a zero-activation no-op referent of 172.3, an unreplicated 3.2 mm not distinguishable from doing nothing, so every finger result for the guidance route is a result about distilling toward a near-no-op target. $n = 12$ seeds throughout. The two routes are not matched in what they consume: the prior arm holds the analytic reward at imagined states, and the Dyna arms train on 64 real and 64 imagined transitions where model-free arms train on 128 real, so equal

update-to-data is not equal real-data signal; the matched arm controls this at withdrawal, not during training. The body-model withdrawal result is an elbow result, and the converged decompositions were run on the elbow only. One withdrawal shape (a step, no taper); seven attachment points give a short trend test, and the loss-term slope is marginal on the elbow and non-monotone on the finger, so we claim that it does not decay, not that it grows monotonically. In the replacing regime *selfanchor* coincides with the teacher by construction, and that regime was run on the elbow only. Contrasts were chosen during the study and not pre-registered. No physical robot is involved, and the humanoid pipelines that motivate the teacher experiments use on-policy learners whose transition costs may differ in size, though not, we expect, in kind.

VII. CONCLUSION

Two routes to a motor prior were compared under one backbone on two musculoskeletal bodies, to a converged budget, with each route given the adversarial control the other has and with the withdrawal of each made single-factor. The body model helps where the body is hard: transient on the elbow, stable on the finger. Guidance is only as good as its teacher: a small lasting advantage where the teacher is competent, a ceiling below model-free learning where it barely moves. And withdrawing either prior costs no dependence: the body model’s cost is a transient the standard protocol misses, and the teacher’s, once a zero-information stand-in separates the loss term from the teacher, never grows with attachment and ends as the loss of a competent teacher. An early endpoint reversed the first two verdicts; matching data budgets changed none.

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